

PUBUDU GUNASEKARA

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MS Computer Science (Northeastern) • Software Engineering Intern | Available Summer 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences - San Jose, CA Jan 2026 - Present
M.S. in Computer Science | GPA: 4.0 / 4.0 | Expected May 2028
Coursework: Algorithms, Programming Design Paradigms, Machine Learning (Fall 2026), Natural Language Processing (Fall 2026)
Conestoga College - Waterloo, ON, Canada 2022 - 2023
Postgraduate Certificate, Mobile Solutions Development | GPA: 3.74 / 4.0
Victoria University - Melbourne, Australia 2018 - 2021
Bachelor of Information Technology | GPA: 6.25 / 7.0

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL
Backend & Distributed: Spring Boot, Node.js / Express, Apache Kafka, Redis, PostgreSQL, MongoDB, REST APIs, hexagonal architecture
Infra & DevOps: Docker, Docker Compose, GitHub Actions, CI/CD, Prometheus, Grafana, AWS
Frontend: React, Next.js, React Native, Vite, Redux
Familiar: C++, Jenkins

EXPERIENCE

Gunasekara Transport - Web Developer Dec 2025 - Jun 2026 · Remote
• Built the corporate website and web app interfaces, plus an internal business management system (HR, operational workflows, and reporting) that replaced manual processes, using React.js, JavaScript, PHP, WordPress, and Docker.
• Managed web infrastructure including domain configuration and email gateway administration.
Virtusa - Associate Engineer, QA Jun 2021 - Mar 2022 · Colombo, Sri Lanka
• Built Java and Selenium WebDriver test automation for a British Telecommunications platform serving 1M+ users, designing and executing test cases and test plans to verify system functionality across agile sprints.
• Led test planning and ran client demos and daily stand-ups for the module under the QA lead's delegation, and collaborated with developers to validate bug fixes.
• Tracked QA metrics including defect densities and open defect counts, working across CST Environment, Camunda Cockpit, RabbitMQ, Swagger UI, and MQ Console.

PROJECTS

Distributed Task Scheduler

Java 21, Spring Boot 3.5, Apache Kafka, Redis, PostgreSQL, Prometheus, Grafana, Docker, GitHub Actions • [Code](#) · [Case Study](#)
• Built a distributed task scheduler on a hexagonal architecture enforced at build time via ArchUnit, with at-least-once delivery over Kafka and a state-machine guard that prevents duplicate execution on redelivery.
• Implemented a Redis distributed lock (atomic SETNX acquire, Lua compare-and-delete release) and an exponential-backoff retry scheduler (10s, 30s, 90s) with dead-letter handling.
• Instrumented the system end-to-end with Prometheus metrics and Grafana dashboards; containerized with Docker Compose and automated build/test in GitHub Actions CI.

AI Code Review Assistant

React 19 / Vite, Node.js / Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS • [Code](#) · [Live Demo](#) · [Case Study](#)
• Built and deployed a full-stack app that reviews a selected GitHub pull request via GitHub OAuth, returning severity- and category-tagged findings from gpt-4o-mini in a strict JSON schema.
• Engineered per-user Redis rate limiting with a fail-open policy and standard rate-limit headers, plus diff truncation to stay within the model's context budget.

Travel Day Scheduler

Python, Weighted A* Search • Graduate algorithms project (team) • [Code](#)
• Co-designed a weighted / anytime A* itinerary planner (team of 2) with iterative weight escalation and heap pruning to bound memory under large search spaces; implemented the initial weighted A* search and built the route-visualization frontend.

ACTIVITIES

• Northeastern University: Husky Code Club (Member) · Graduate Student Career Advisory Board (Founding Cohort Member)